Resident Study: Automated Feedback in VR Mastoidectomy

A small user study comparing two forms of automated feedback after a virtual reality temporal bone drilling session. Your performance is not a primary endpoint and will not be used to evaluate you.

What you will do (about 30–40 minutes)

1. **Tutorial.** In the headset, a short guided tutorial on the drill, haptic device, microscope, zoom, foot pedals and pause menu. Practice bone is restored afterwards.
2. **Timed trial.** 10 minutes of drilling time to perform a cortical mastoidectomy and identify the short process of the incus. The timer starts at first bone contact and stops while the pause menu is open. It is fine not to finish. If you finish early, hold the left foot pedal and select **End trial**.
3. **Questionnaires and feedback.** On the desktop:
    - a brief background questionnaire and a rating of your own performance;
    - **Video-Based Feedback**: selected moments from your session video, compared with experts;
    - **Metric-Based Feedback**: your safety and drilling-motion metrics for each phase, compared with experts;
    - a short questionnaire after each format, then which format you preferred and a brief reflection.

The two feedback formats are shown in alternating order across participants.